COMPUTER PROGRAMMING

LAB 3

LAB TITLE

Relational Operators, Logical Operators and Conditional Statements in C++

1. OBJECTIVE

The objective of this lab is to understand how relational operators, logical operators, and conditional statements work in C++. In this lab we learn how to compare values, evaluate logical conditions, and control program flow using if, if-else, and else-if statements.

2. THEORY

In C++ programs normally execute instructions one after another. However, sometimes we need to make decisions depending on certain conditions. For this purpose C++ provides relational operators, logical operators, and conditional statements.

Relational Operators

Relational operators are used to compare two values.

Operator	Meaning
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
==	Equal to
!=	Not equal to

Logical Operators

Logical operators are used to combine multiple conditions.

Operator	Meaning
&&	Logical AND
||	Logical OR
!	Logical NOT

Conditional Statements

Conditional statements allow a program to execute different blocks of code depending on conditions.

Types of conditional statements:

- **if statement**
- **if–else statement**
- **else–if statement**

3. SOFTWARE

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

PROGRAM 1

Comparison Using Relational Operators

Algorithm

- Start the program
- Declare two integer variables
- Take two numbers from the user
- Compare the numbers using relational operators
- Display the comparison result
- End the program

```
#include <iostream>
using namespace std;

int main()
{
   int a, b;

   cout << "Enter two numbers: ";
   cin >> a >> b;

   if(a == b)
      cout << "Both numbers are equal";

   if(a > b)
      cout << "First number is greater";

   if(a < b)
      cout << "First number is smaller";

   return 0;
}
```

PROGRAM 2

Logical Operators

Algorithm

- Start the program
- Declare an integer variable
- Take a number from the user
- Check if the number is greater than 0 and less than 100
- Display the result
- End the program

```
   #include <iostream>
using namespace std;
int main()
{
int num;
 cout << "Enter a number: ";

   cin >> num;
if(num > 0 && num < 100)

  cout << "Number is between 0 and 100";

   else

       cout << "Number is not between 0 and 100";
  return 0;
}
```

PROGRAM 2

If Statement (Positive Number)

Algorithm

1. Start the program
2. Declare an integer variable
3. Take a number from the user
4. Check if the number is positive
5. Display the message
6. End the program

```
#include <iostream>

using namespace std;

int main()

{   int num;

  cout << "Enter a number: ";
```

```
  cin >> num;
 if(num > 0)

      cout << "The number is positive";

 return 0;

}
```

Conclusion

In this lab we learned how to use relational and logical operators in C++. We also learned how conditional statements such as **if**, **if-else**, and **else-if** help a program make decisions based on different conditions. These concepts are very important for writing logical programs.